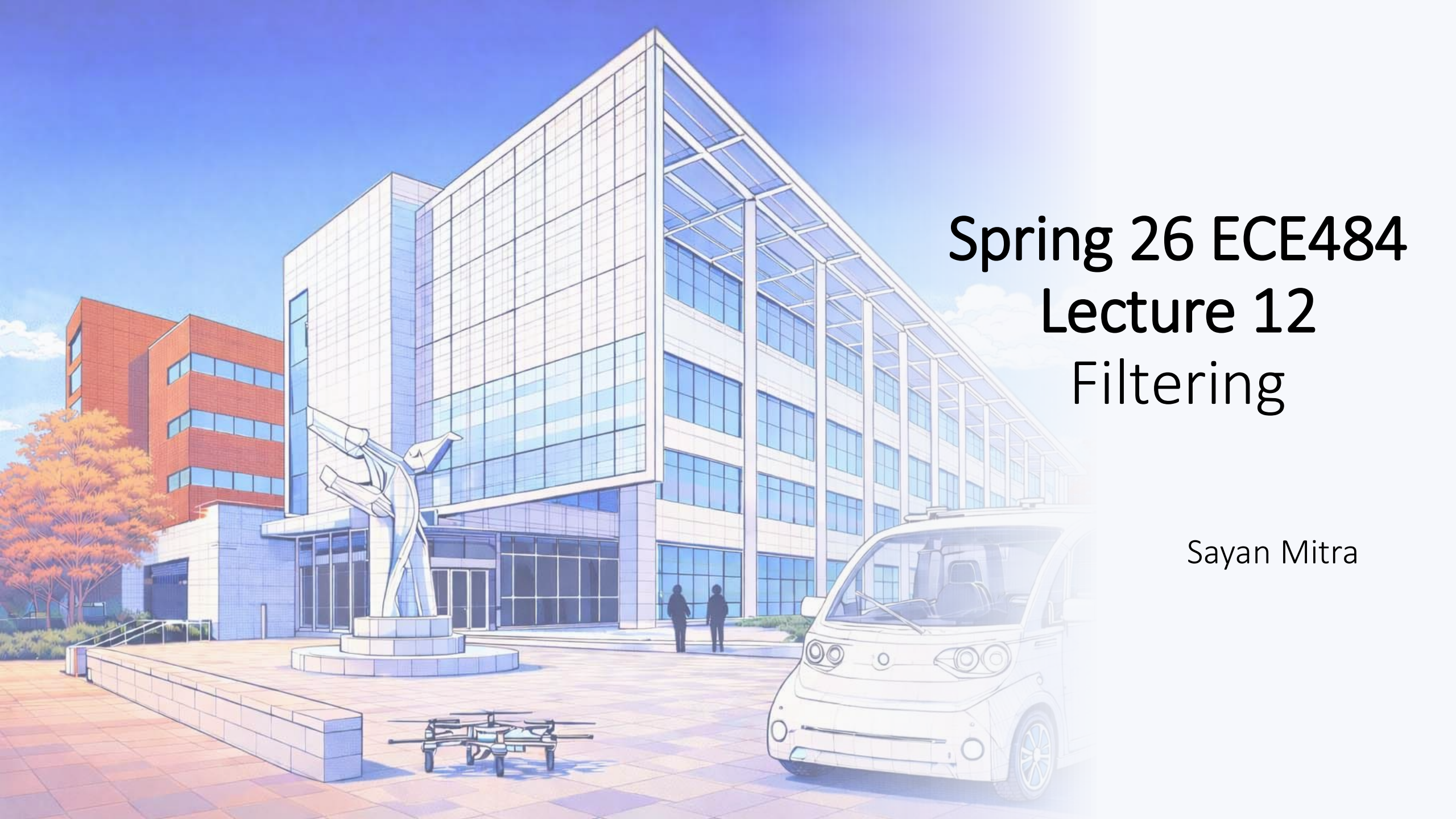




Spring 26 ECE484 Lecture 12 Filtering

Sayan Mitra

Finding a Needle in the Atlantic Ocean

*May 1968 — A nuclear submarine vanishes with 99 souls aboard.
The U.S. Navy has no idea where — or why.*

The USS Scorpion — a nuclear attack submarine — disappeared in a 2,600-mile stretch of open ocean. Dr. John Craven led the search using Bayesian estimation:

1. Hypothesize — interview experts to get prior $P(\text{failure scenario } j)$ for 9 scenarios ($j = \text{fire, torpedo malfunction, Soviet attack...}$)
2. Model— divide the ocean into grid squares, assign $P(\text{wreck in grid } i \mid \text{scenario } j)$ using physics
3. Total probability

$$P(\text{wreck in } i) = \sum_j P(\text{wreck in square } i \mid \text{scenario } j) \cdot P(\text{scenario } j)$$

**Result: Found 220 yards from predicted location — under 2 miles of water.
Bayesian estimation found Scorpion. The Kalman filter IS Bayesian estimation.**

Today's lecture Bayes' filter

The State Estimation / Filtering Problem

Goal: Recursively estimate state x_t from noisy measurements $z_{1:t}$ & controls $u_{1:t}$

Compute the **belief**: $\text{bel}(x_t) = p(x_t \mid z_{1:t}, u_{1:t})$

x_t — State

The true state at time t

E.g. robot pose (x, y, θ)

u_t — Control

Control applied

to transition from $t-1$ to t

E.g. velocity command

z_t — Measurement

Observation received

at time t

E.g. LIDAR range scan

Motion model: $p_D(x_t \mid x_{t-1}, u_t)$

Measurement model: $p_M(z_t \mid x_t)$

Review: Markov Decision Processes (MDP)

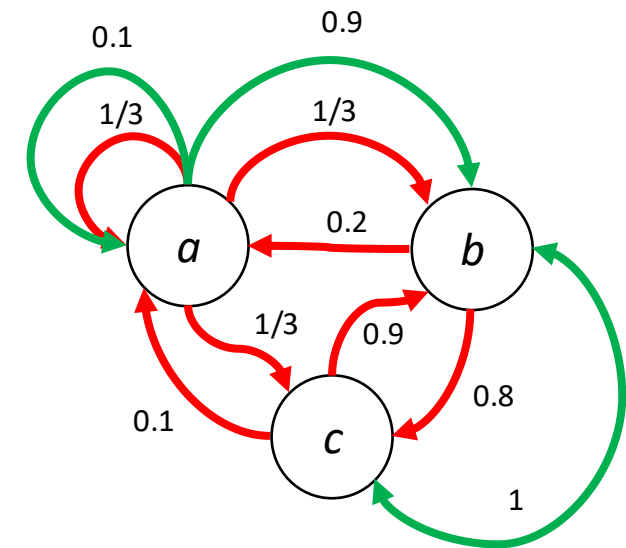
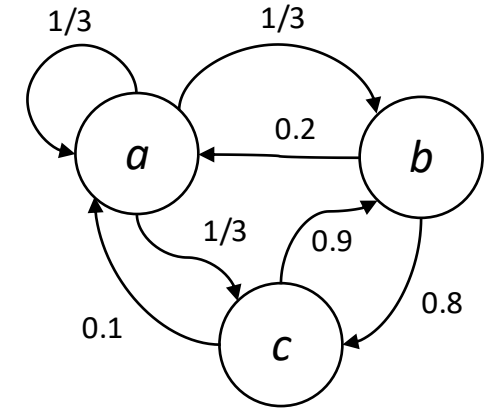
$$p_D(X_{t+1} = b | X_t = a, U_t = \text{red}) = \frac{1}{3}$$

p_D represented by a collection of matrices $\mathbf{D}: U \rightarrow \mathbb{R}^{|Q| \times |Q|}$
where $D_{ij}(u)$ is the probability of state i to transition to j under action u

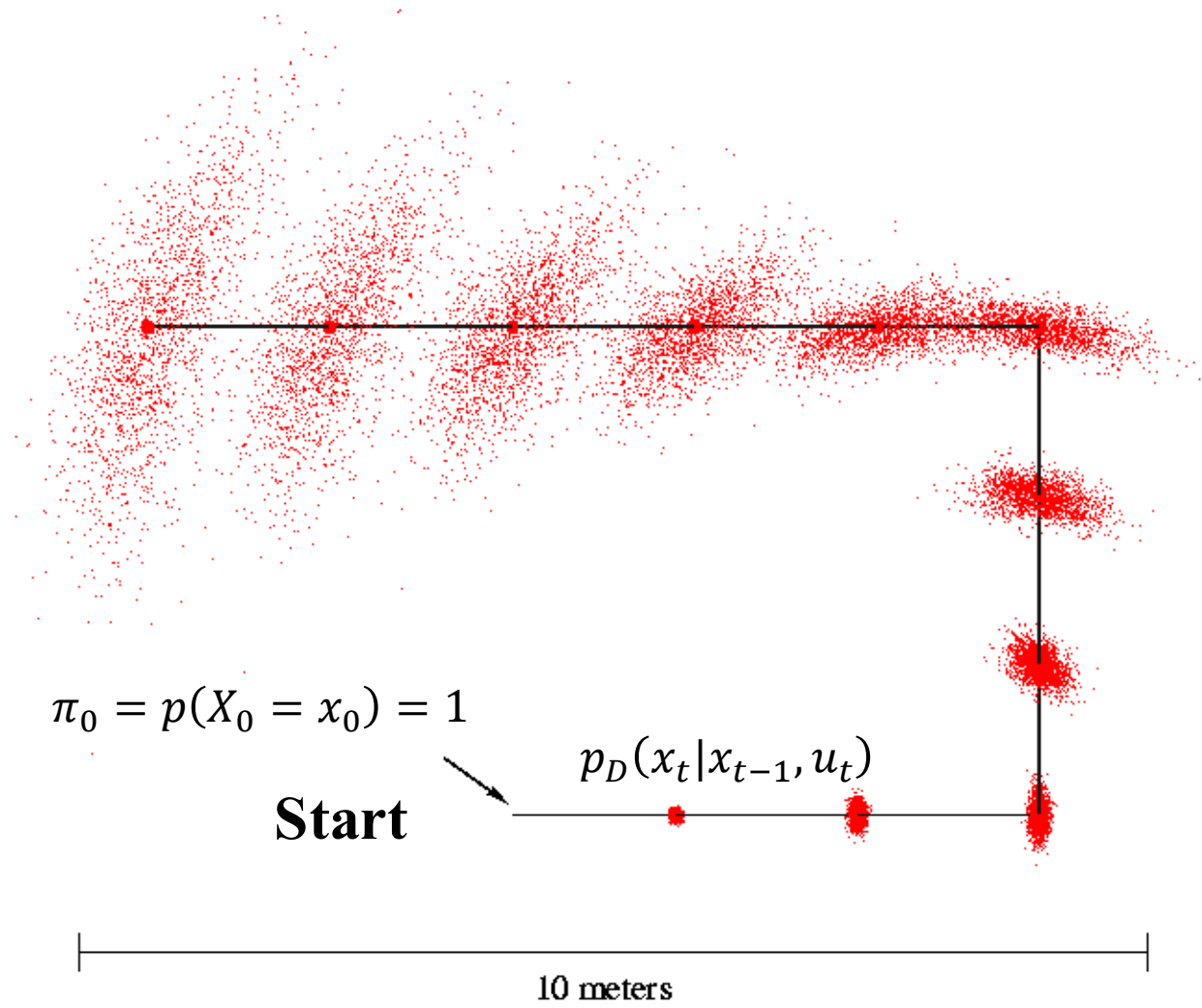
$$\begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{4}{5} \\ \frac{1}{10} & \frac{9}{10} & 0 \end{bmatrix} \quad \begin{bmatrix} \frac{1}{10} & \frac{9}{10} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In general the transition probabilities can be time varying

$p_D(x'|x, u)$ if transition probabilities are time invariant



Example Motion Model without measurements



The state transition probabilities are defined by
 $x_{t+1} = f(x_t, u_t) + \omega_t$

where $\omega_t \sim N(0,1)$

Probabilistic measurements

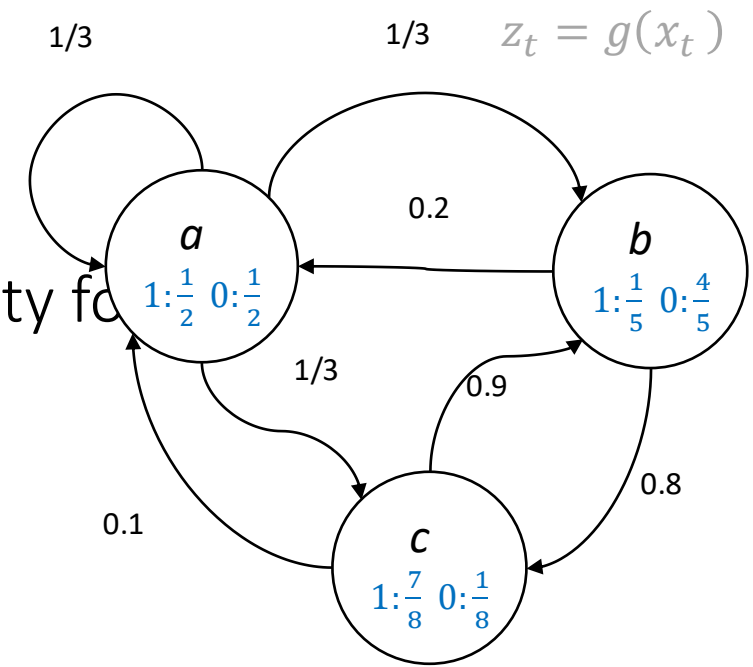
Measurement model gives the output/observation probability for a given state

$$p_M(z_t = 1 | x_t = a) = \frac{1}{2}$$

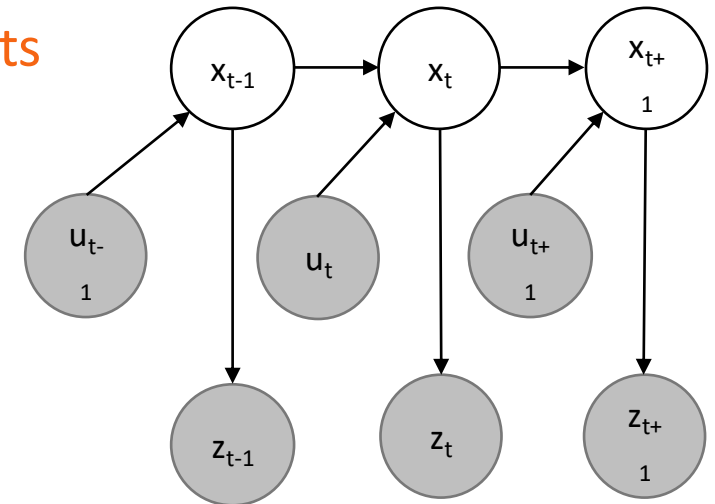
Generally, measurements can depend on history

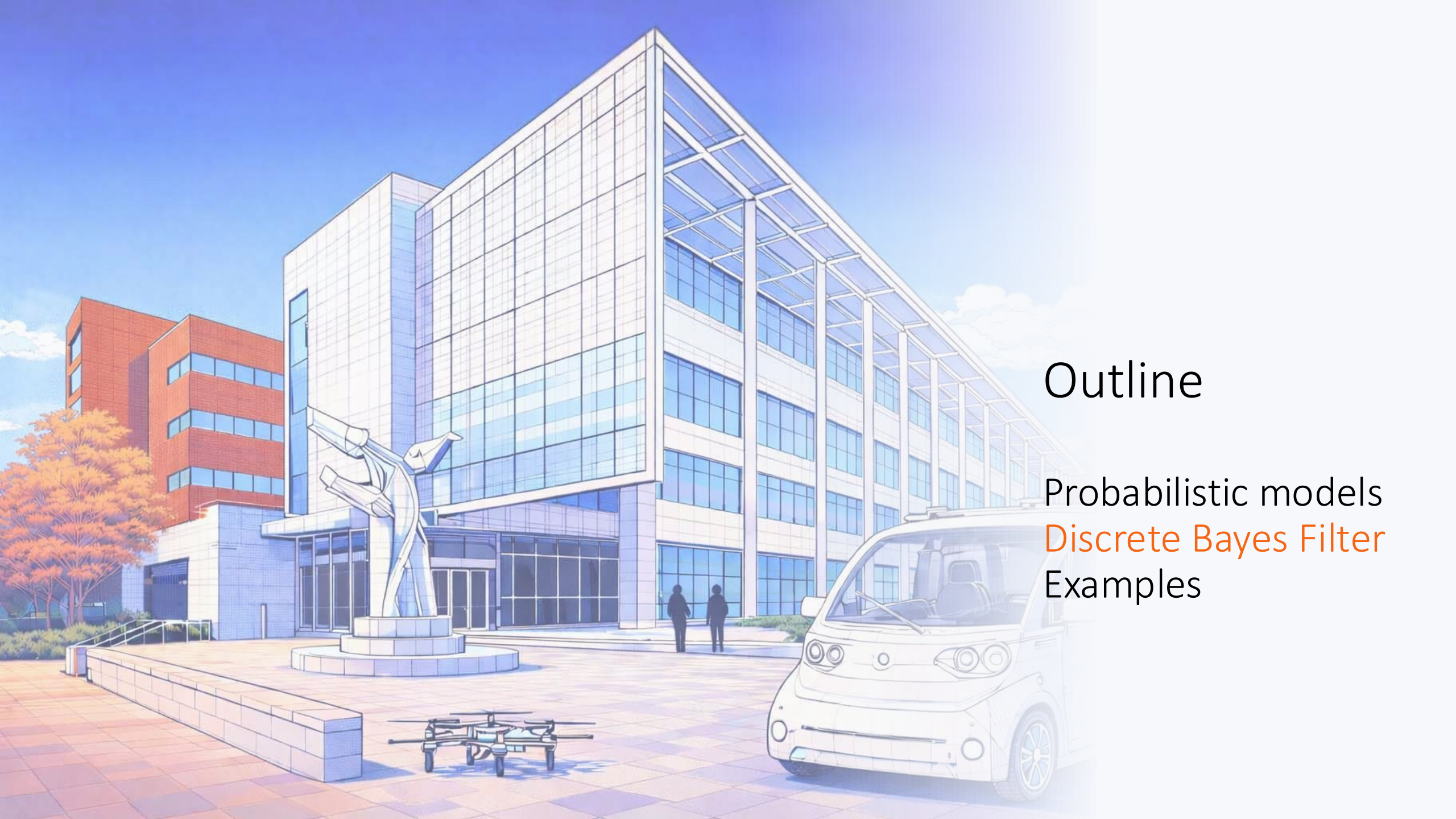
$$p_M(z_t | x_{0:t}, z_{1:t-1}, u_{0:t-1})$$

- If **state is complete or conditional independence of measurements**
- $p_M(z_t | x_{0:t}, z_{1:t-1}, u_{1:t}) = p(z_t | x_t)$
- $p_M(z_t | x_t)$: measurement probability
- $p_M(z | x)$: **time invariant measurement probability**



State a produces output 1 and 0 each with probability 0.5





Outline

Probabilistic models
Discrete Bayes Filter
Examples

Step 1: Apply Bayes' Rule (Update)

Start with the definition of belief:

$$\begin{aligned} \text{bel}(x_t) &= p(x_t | z_{1:t}, u_{1:t}) = p(x_t | z_t, z_{1:t-1}, u_{1:t}) \\ &= \eta \cdot p(z_t | x_t, z_{1:t-1}, u_{1:t}) \cdot p(x_t | z_{1:t-1}, u_{1:t}) \end{aligned}$$

[Applying Bayes' rule with z_t where η is the normalizer]

$$= \eta \cdot p_M(z_t | x_t) \cdot p(x_t | z_{1:t-1}, u_{1:t})$$

[Applying cond. independence $p(z_t | x_t, z_{1:t-1}, u_{1:t}) =$

$$p_M(z_t | x_t)]$$

$$\text{bel}(x_t) = \eta \cdot p_M(z_t | x_t) \overline{\text{bel}}(x_t) \quad \textbf{where} \quad \overline{\text{bel}}(x_t) = p(x_t | z_{1:t-1}, u_{1:t})$$

Step 2: Total Probability (Prediction)

How do we compute $\overline{bel}(x_t)$? Marginalize over x_{t-1} or Law of **total probability**

$$\begin{aligned}\overline{bel}(x_t) &= p(x_t \mid z_{1:t-1}, u_{1:t}) \\ &= \sum_{x \in Q} p_D(x_t \mid \mathbf{x}_{t-1}, z_{1:t-1}, u_{1:t}) bel(\mathbf{x}_{t-1} \mid z_{1:t-1}, u_{1:t}) \\ &= \sum_{x \in Q} p_D(x_t \mid \mathbf{x}_{t-1}, u_{1:t}) bel(\mathbf{x}_{t-1} \mid z_{1:t-1}, u_{1:t}) \text{ [Applying Markov property]}\end{aligned}$$

$$\overline{bel}(x_t) = \sum_{x \in Q} p_D(x_t \mid \mathbf{x}_{t-1}, u_{1:t}) bel(\mathbf{x}_{t-1} \mid z_{1:t-1}, u_{1:t})$$

The Bayes Filter Algorithm

BAYES_FILTER($bel(x_{t-1}), u_t, z_t$)

PREDICT

for all x_t **do**:

$$\overline{bel}(x_t) = \sum_{x \in Q} p_D(x_t | x_{t-1}, u_t) bel(x_{t-1})$$

UPDATE

$$bel(x_t) = \eta \cdot p_M(z_t | x_t) \overline{bel}(x_t)$$

endfor

return $bel(x_t)$

Key Properties

- Recursive: only needs $bel(x_{t-1})$, not the full history
- General: works for any state-space model
- Problem: the sum/integral is often intractable in closed form!

Bayes Filter: Continuous Distributions

Bayes_filter($bel(x_{t-1}), u_t, z_t$)

for all x_t do:

$$\overline{bel}(x_t) = \int p(x_t | u_t, x_{t-1}) bel(x_{t-1}) dx_{t-1}$$

$$bel(x_t) = \eta p(z_t | x_t) \overline{bel}(x_t)$$

end for

return $bel(x_t)$

Example: Measurement Model p_M

Discrete Gaussian sensor model. Given the drone is truly at x , the sensor reading z_k is distributed as:

$$p_M(z_k | x) \propto \exp\left(-\frac{(x - z_k)^2}{2\sigma^2}\right), \sigma = 1 \text{ m}$$

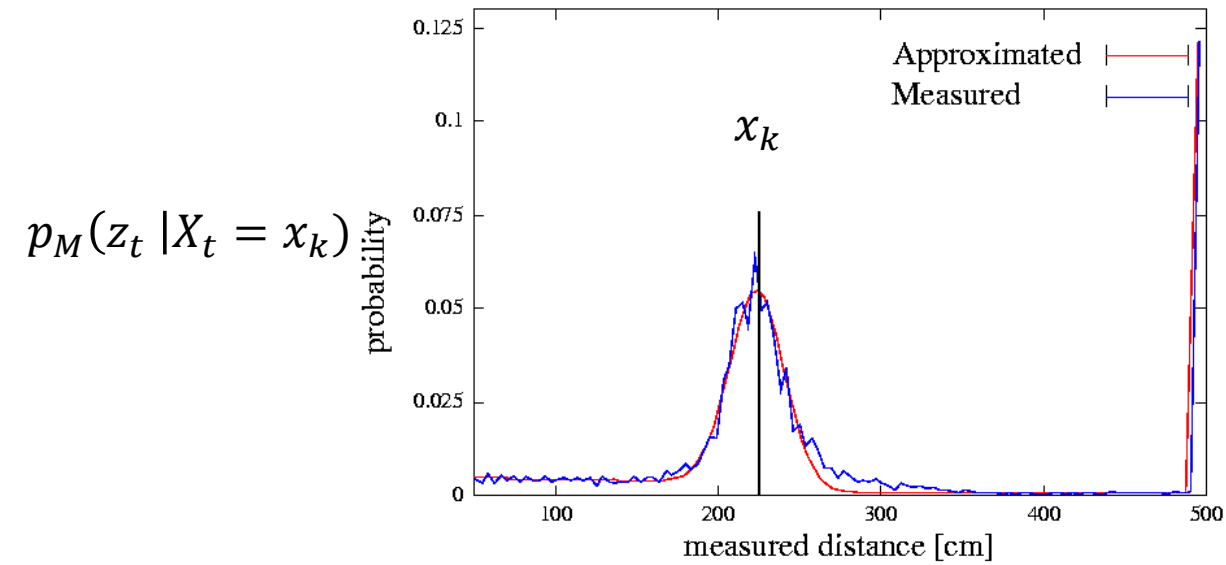
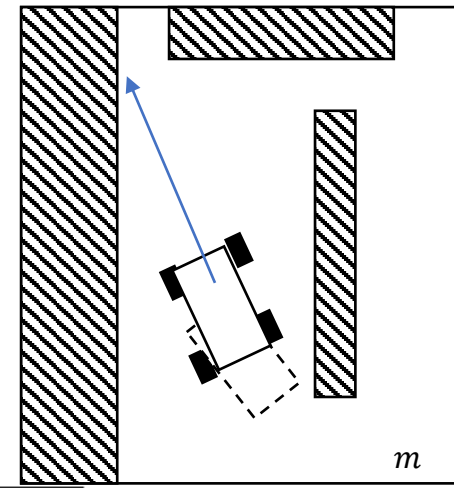
The likelihood vector $\ell(x) = p_M(z_k | x)$ evaluated at each state — given a sensor reading — tells us how well each state explains that reading.

Normalized likelihood values for $z_k = +1 \text{ m}$:

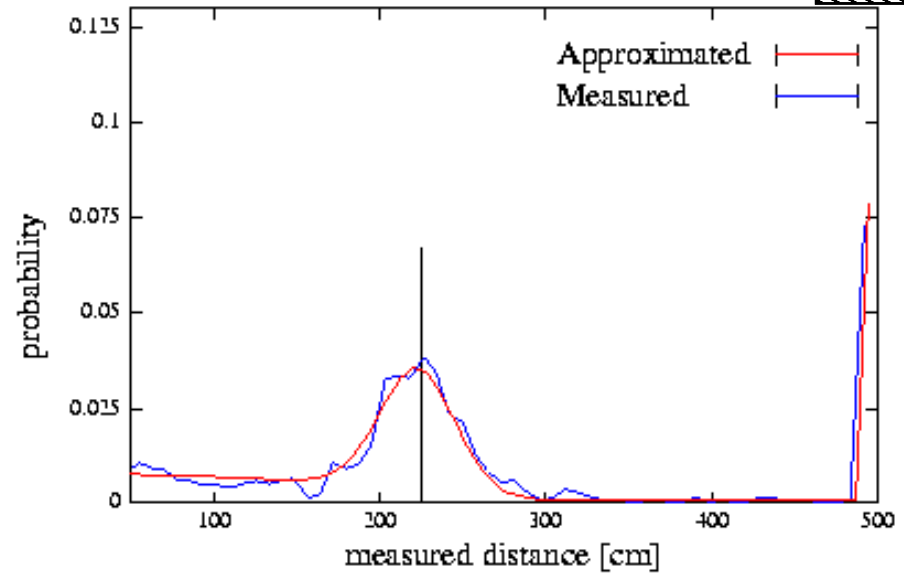
x	-2	-1	0	+1	+2
p_M	0.02	0.08	0.25	0.45	0.20

Peak at $x = +1$ (matches z_k); falls off for states farther away. Slight asymmetry (0.25 vs 0.20) is from discrete normalization.

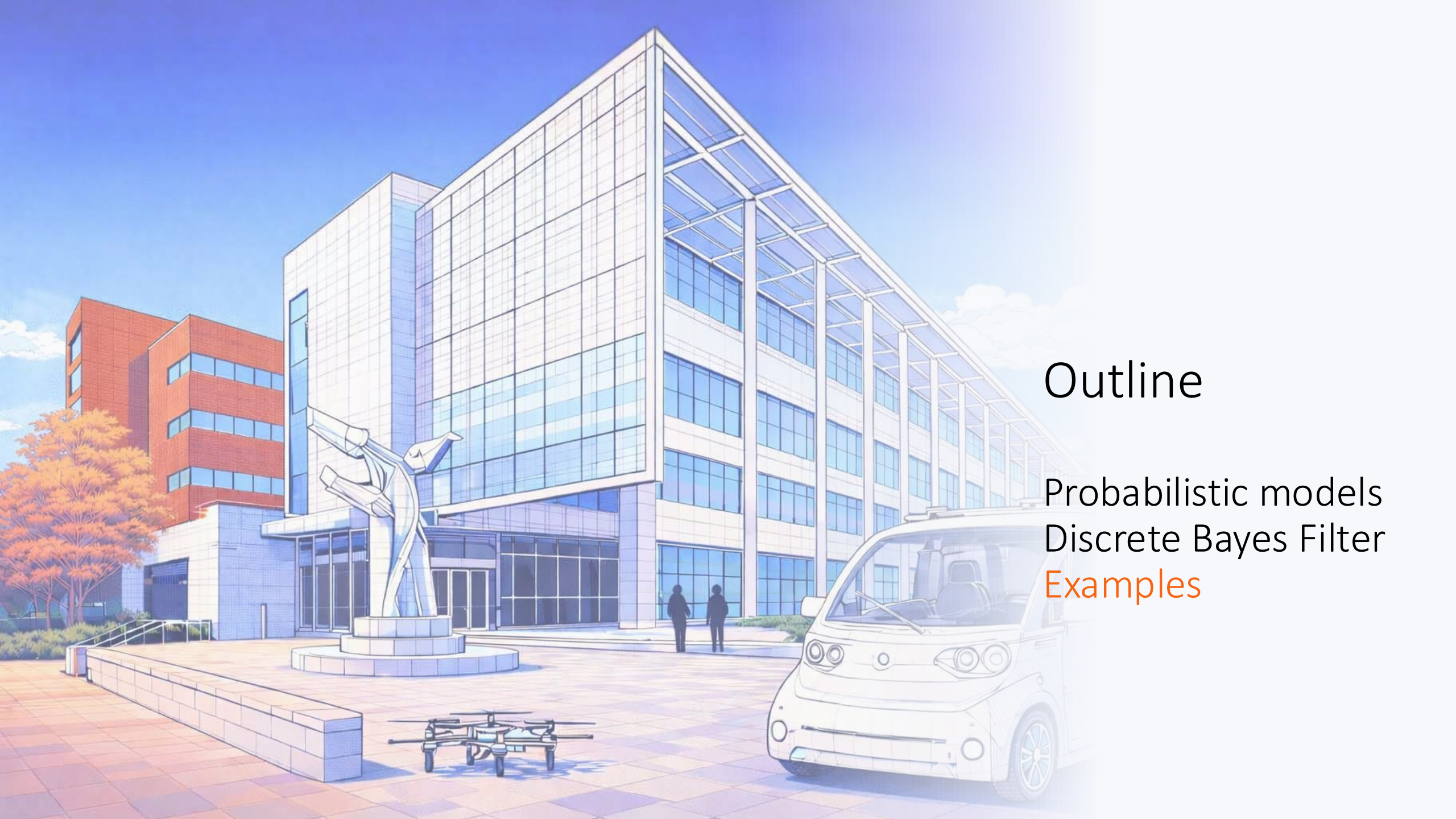
Example Proximity Sensor Measurement Models



Laser sensor



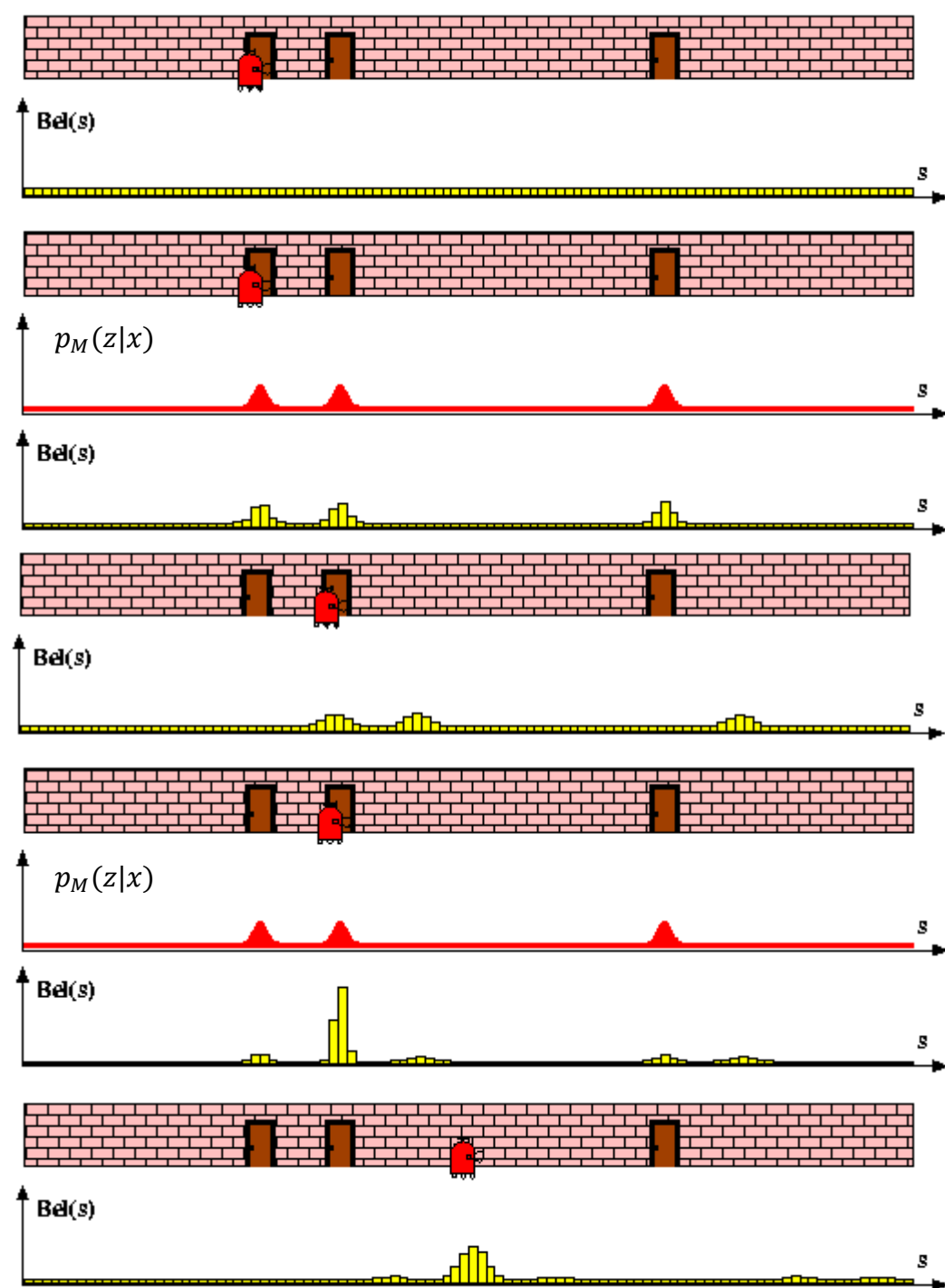
Sonar sensor



Outline

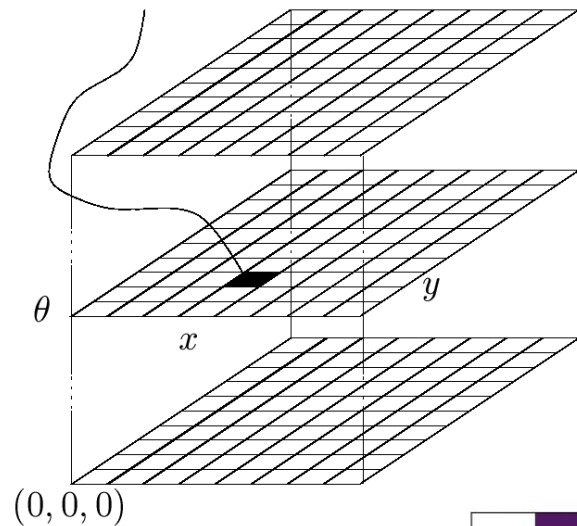
Probabilistic models
Discrete Bayes Filter
Examples

Grid localization,
 $bel(x_t)$ represented by a
 histogram over grid



Grid localization with bicycle model + landmarks

$$\text{bel}(X_t = \langle x, y, \theta \rangle)$$

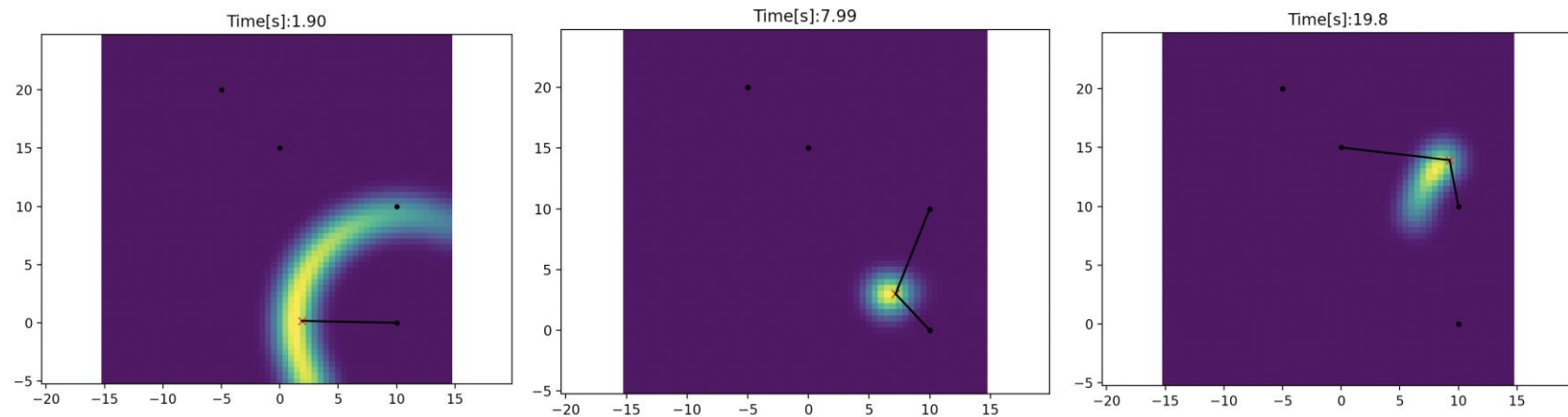


The state space Q is a quantization of position and orientation $q = \langle x, y, \theta \rangle$

A belief is a probability distribution over states $\text{bel}(q_t) \in P(Q)$

Prediction: Fixing an (steering) input u_t compute the new intermediate belief over Q using motion model $p_D(q_{t+1}|q_t, u_{t+1})$

Correction: Update intermediate belief with received distance to landmark z_{t+1} based on measurement model p_M



Histogram Filter Example — Iteration 1

States: $Q = \{-2, -1, 0, +1, +2\}$ True $x_0 = -2$ $\text{bel}(x_0) = [0.20, 0.20, 0.20, 0.20, 0.20]$ (uniform)

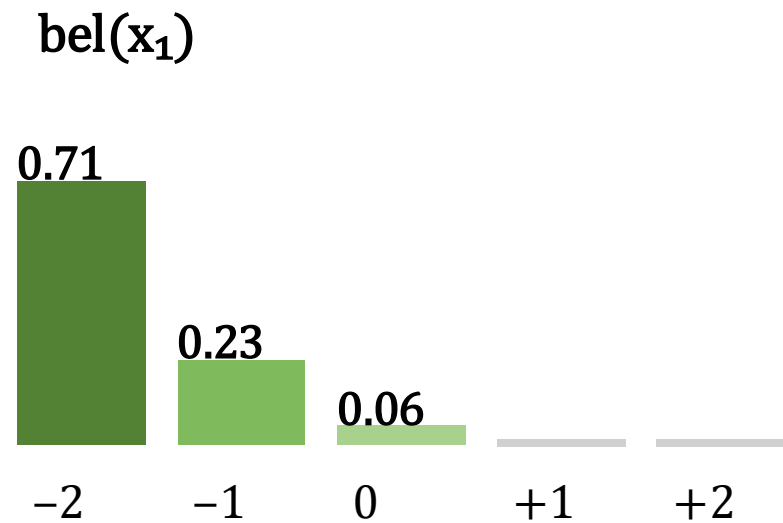
Step 1a — Prediction: $\text{bel}(x_1) = \sum P(x_1 \mid x_0) \cdot \text{bel}(x_0)$

Since $\text{bel}(x_0)$ is uniform, the prediction stays uniform:
 $\text{bel}(x_1) = [0.20, 0.20, 0.20, 0.20, 0.20]$

Step 1b — Update: observe $z_1 = -2$ (true $x_1 = -2$)

$$\text{bel}(x_1) = \eta \cdot p_m(z_1 \mid x_1) \cdot \text{bel}(x_1)$$

x_1	-2	-1	0	+1	+2
$p_m(z_1=-2 \mid x_1)$	0.25	0.08	0.02	0.00	0.00
$\text{bel}(x_1)$	0.20	0.20	0.20	0.20	0.20
Unnormalized	0.050	0.016	0.004	0.000	0.000
$\text{bel}(x_1)$	0.714	0.229	0.057	0.000	0.000



→ After $z_1 = -2$: $\text{bel}(x_1)$ peaks at $x_1 = -2$ with 71.4% — correctly identifies the true state!

$$\eta = 1/0.070 \approx 14.29$$

Histogram Filter Example — Iteration 2

Step 2a — Prediction: $\text{bel}(x_2) = \sum P(x_2 \mid x_1) \cdot \text{bel}(x_1)$

$\text{bel}(x_2) = T^T \cdot \text{bel}(x_1) = [0.617, 0.291, 0.080, 0.011, 0.000]$
(Transition matrix spreads probability to neighboring states → increased uncertainty)

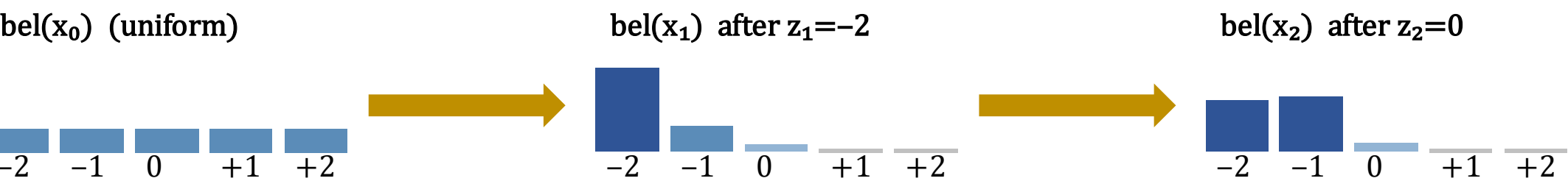
Step 2b — Update: observe $z_2 = 0$ (true $x_2 = -1$)

$$\text{bel}(x_2) = \eta \cdot p_m(z_2 \mid x_2) \cdot \text{bel}(x_2)$$

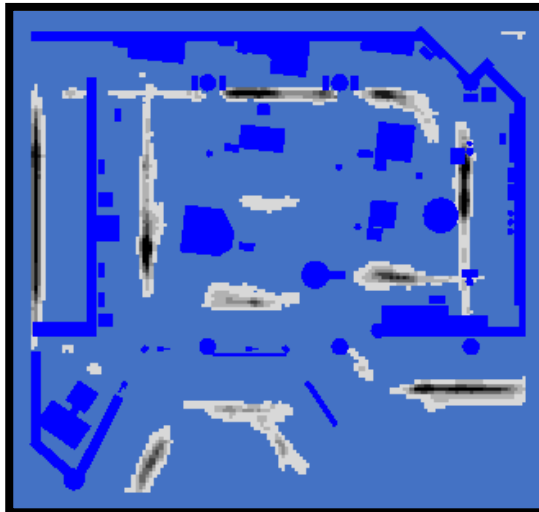
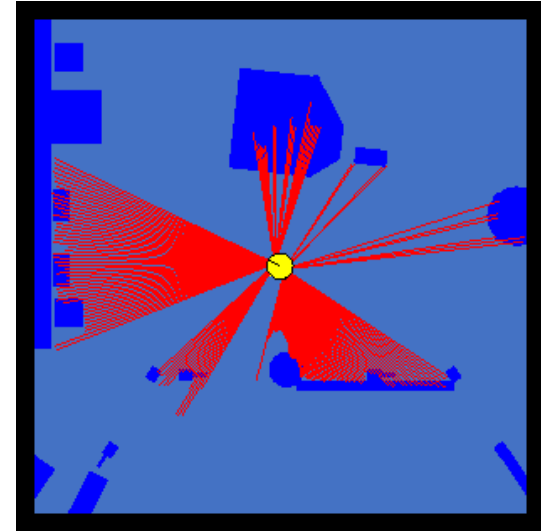
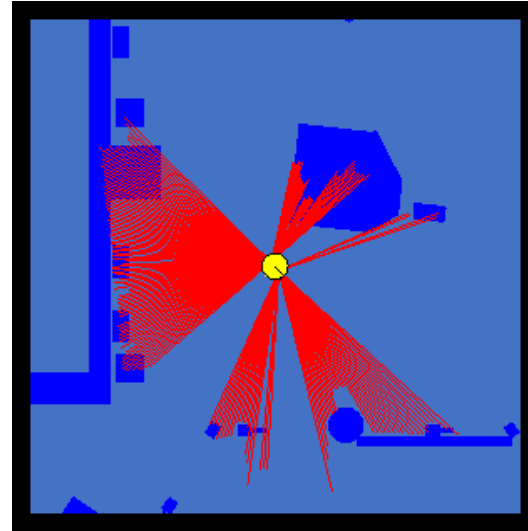
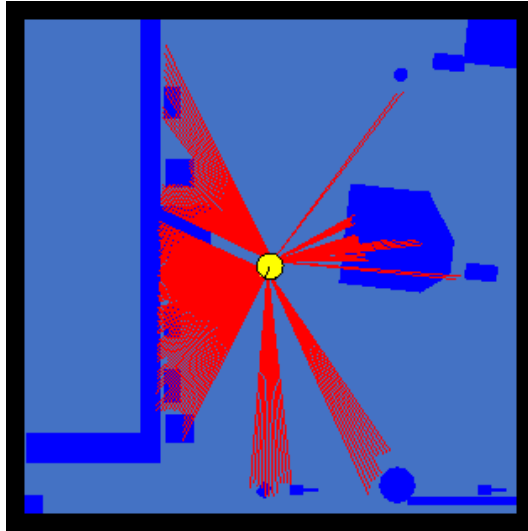
x_2	-2	-1	0	+1	+2
$p_m(z_2=0 \mid x_2)$	0.20	0.45	0.25	0.08	0.02
$\text{bel}(x_2)$	0.617	0.291	0.080	0.011	0.000
Unnormalized	0.123	0.131	0.020	0.001	0.000
$\text{bel}(x_2)$	0.448	0.476	0.073	0.003	0.000

→ $\text{bel}(x_2)$ peaks at $x_2 = -1$ with 47.6% — tracks the shift!

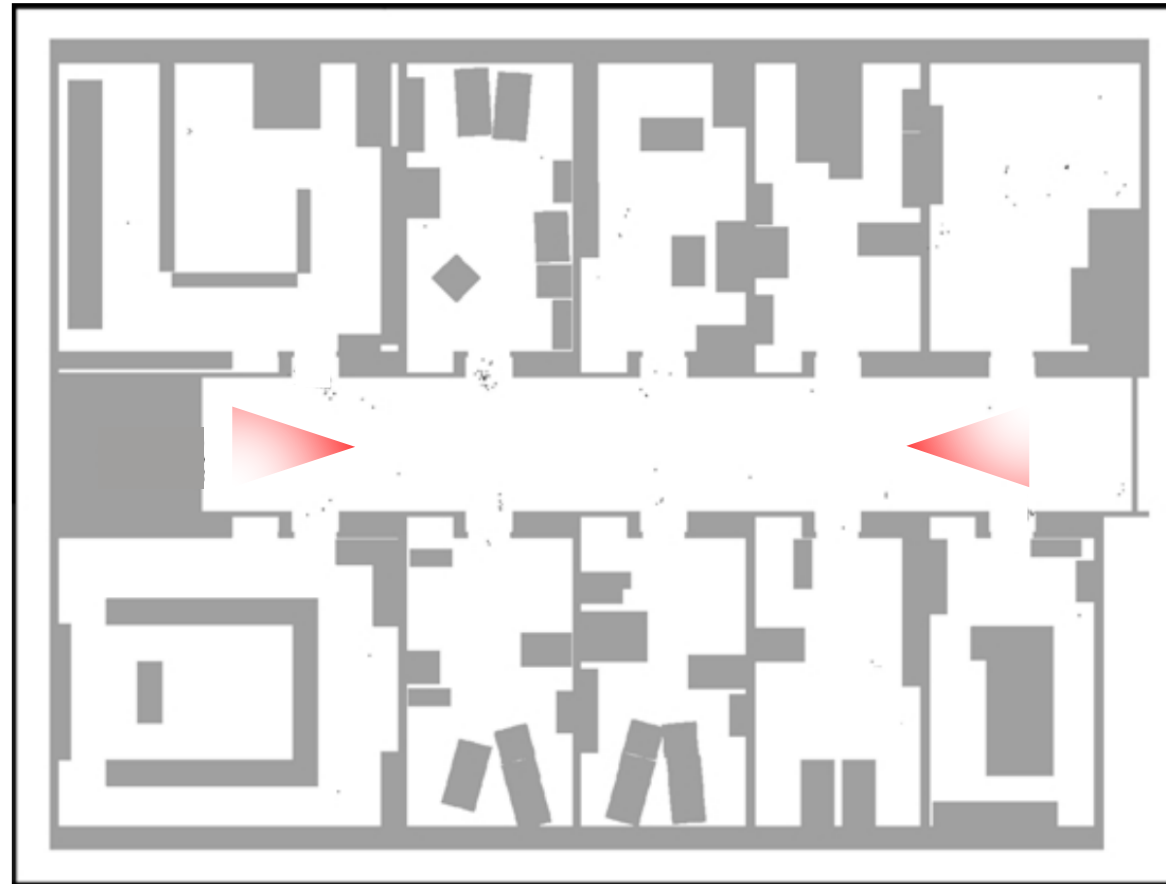
Belief Convergence — How the Histogram Peaks



Grid-based Localization



Ambiguity in global localization arising from locally symmetric environment



Optimality of Bayes Filter

MLE (Maximum Likelihood Estimate)

$$\hat{x}_t^{\text{MLE}} = \operatorname{argmax}_x p(z_t | x)$$

Finds the state that best explains the measurement alone — no prior.

It does not use $\text{bel}(x_t)$ at all, so it ignores all history.

MAP (Maximum A Posteriori)

$$\hat{x}_t^{\text{MAP}} = \operatorname{argmax}_x \text{bel}(x_t) = \operatorname{argmax}_x p(x_t | z_{1:t}, u_{1:t})$$

Finds the mode of the posterior. Optimal under 0–1 loss:

$$L(\hat{x}, x) = 0 \text{ if } \hat{x} = x, \quad 1 \text{ if } \hat{x} \neq x$$

MMSE (Minimum Mean Square Error)

$$\hat{x}_t^{\text{MMSE}} = E[x_t | z_{1:t}, u_{1:t}] = \int x \text{bel}(x_t) dx$$

Finds the mean of the posterior. Optimal under squared error loss:

$$L(\hat{x}, x) = (\hat{x} - x)^2$$

Discrete Bayes (Histogram) Filter — MLE, MAP, and MMSE

MLE: Not directly computed. The filter maintains $\text{bel}(x_t)$, not $p(z_t | x)$.
However, the measurement update step uses likelihoods internally.

MAP: $\hat{x}_t^{\text{MAP}} = \text{argmax}_x \text{bel}(x_t)$. Simply pick the grid cell with highest belief.
This is the most common point estimate from a histogram filter.

MMSE: $\hat{x}_t^{\text{MMSE}} = \sum_x x \cdot \text{bel}(x_t)$. Computed as a weighted average over the grid.
Unlike the KF, $\text{MAP} \neq \text{MMSE}$ in general (the posterior can be multimodal).

Key Properties of Bayes Filter

Nonparametric: No assumption on the shape of $\text{bel}(x_t)$ — it can be multimodal, skewed, or arbitrary.

Exact (on the grid): $\text{bel}(x_t)$ is the true Bayesian posterior over the discrete state space — no distributional approximation (unlike the KF, which forces a Gaussian). Therefore MAP and MMSE derived from $\text{bel}(x_t)$ are optimal within the grid. The only source of error is grid resolution, not the algorithm.

Computational cost: $O(|Q|^2)$ per step (prediction requires summing over all state pairs). Scales poorly to high-dimensional state spaces.

No optimality guarantee: Unlike the KF, the histogram filter does not minimize any global error criterion. It is a consistent estimator as $\Delta x \rightarrow 0$ but is not MVUE or BLUE in general.

From Bayes Filter to Kalman Filter

The Problem: The integral $\text{bel}(x_t) = \int pD(x_t \mid x_{t-1}, u_t) \cdot \text{bel}(x_{t-1}) dx_{t-1}$ is generally intractable. What if we restrict the distributions?

① Linear Models

Motion: $x_t = A_t x_{t-1} + B_t u_t + \varepsilon_t$

Measurement: $z_t = C_t x_t + \delta_t$

$\varepsilon_t \sim N(0, R_t)$ $\delta_t \sim N(0, Q_t)$

② Gaussian Distributions

Initial belief is Gaussian:

$\text{bel}(x_0) = N(\mu_0, \Sigma_0)$

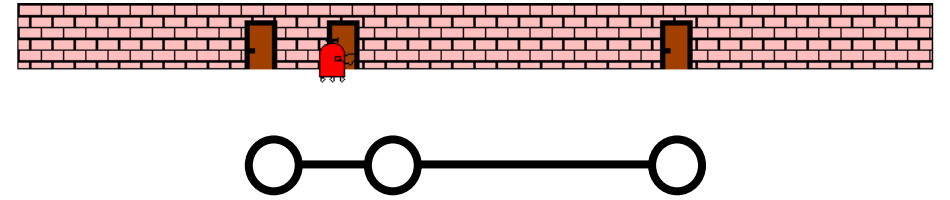
Gaussians are closed under

KEY RESULT: If $\text{bel}(x_{t-1})$ is Gaussian and models are linear-Gaussian, then $\text{bel}(x_t)$ and $\text{bel}(x_t)$ are also Gaussian.

We only need to track μ_t and Σ_t .

Next: Derive closed-form predict and update for μ_t and Σ_t → The Kalman Filter!

Summary



- Key variable: Grid resolution
- Two approaches
 - Topological: break-up pose space into regions of significance (landmarks)
 - Metric: fine-grained uniform partitioning; more accurate at the expense of higher computation costs
- Important to compensate for coarseness of resolution
 - Evaluating measurement/motion based on the center of the region may not be enough. *If motion is updated every 1s, robot moves at 10 cm/s, and the grid resolution is 1m, then naïve implementation will not have any state transition!*
- Computation
 - Motion model update for a 3D grid required a 6D operation, measurement update 3D
 - With fine-grained models, the algorithm cannot be run in real-time
 - Some calculations can be cached (ray-casting results)

Extra slides

Beliefs

Belief: Robot's knowledge about the state

True state x_t is not measurable robot must estimate state $\hat{x}_t$ from measurements and this in Bayesian filtering the estimate is the *belief*

$$bel(x_t) = p(x_t | z_{1:t}, u_{1:t}) \in \mathcal{P}(Q)$$

Posterior over states at time t given all past measurements and control.

Calculated in two steps:

Initially: $bel(x_0) = \pi_0$

1. **Prediction**: $\overline{bel}(x_t) = p(x_t | \mathbf{z}_{1:t-1}, u_{1:t})$ based *all but last* measurements & all control
2. **Correction**: $bel(x_t)$ from $\overline{bel}(x_t)$ based on last measurement $\mathbf{z}_t$

Bayes Filter: Prediction and Correction

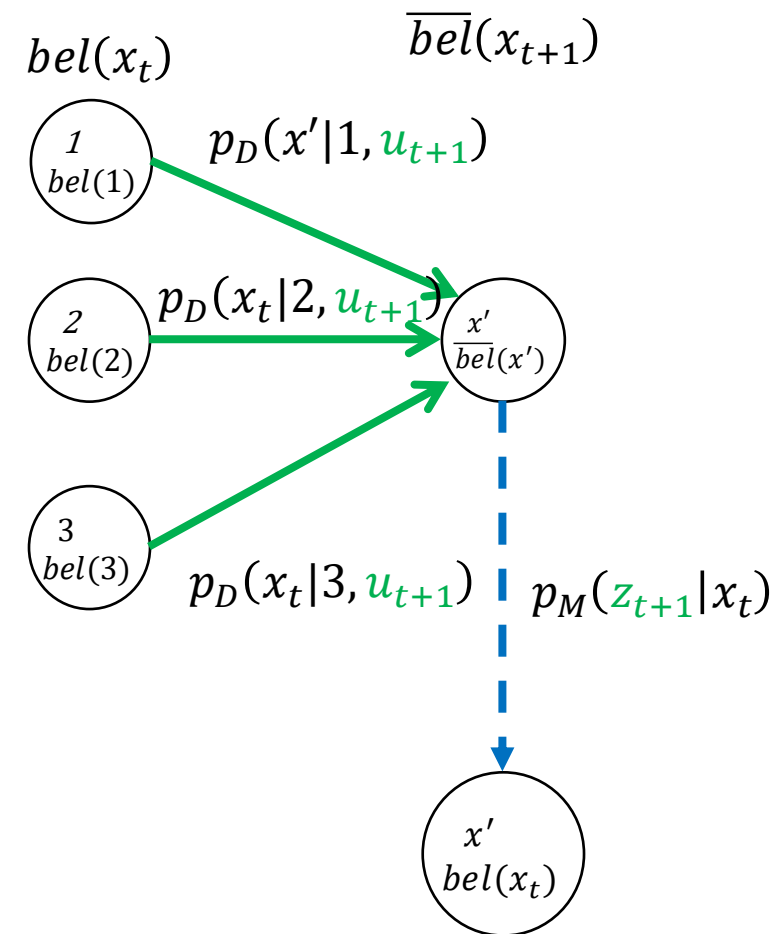
Bayes_filter($bel(x_t), u_{t+1}, z_{t+1}$) calculates $bel(x_{t+1})$

$$\overline{bel}(x_{t+1}) = p(x_{t+1} | \mathbf{z}_{1:t}, u_{1:t+1}) = p(x_{t+1} | u_{t+1})$$

intermediate belief without $\mathbf{z}_{1:t+1}$

For discrete distributions for each $x' \in Q$

$$\overline{bel}(X_{t+1} = x') = \sum_{x \in Q} p_D(X_{t+1} = x' | X_t = x, U_{t+1} = u_{t+1}) bel(X_t = x)$$



Bayes Filter: Prediction and Correction

Bayes_filter($bel(x_t), u_{t+1}, z_{t+1}$) calculates $bel(x_{t+1})$

$$\overline{bel}(X_{t+1} = x') = \sum_{x \in Q} p_D(X_{t+1} = x' | X_t = x, U_{t+1} = u_{t+1}) bel(X_t = x)$$

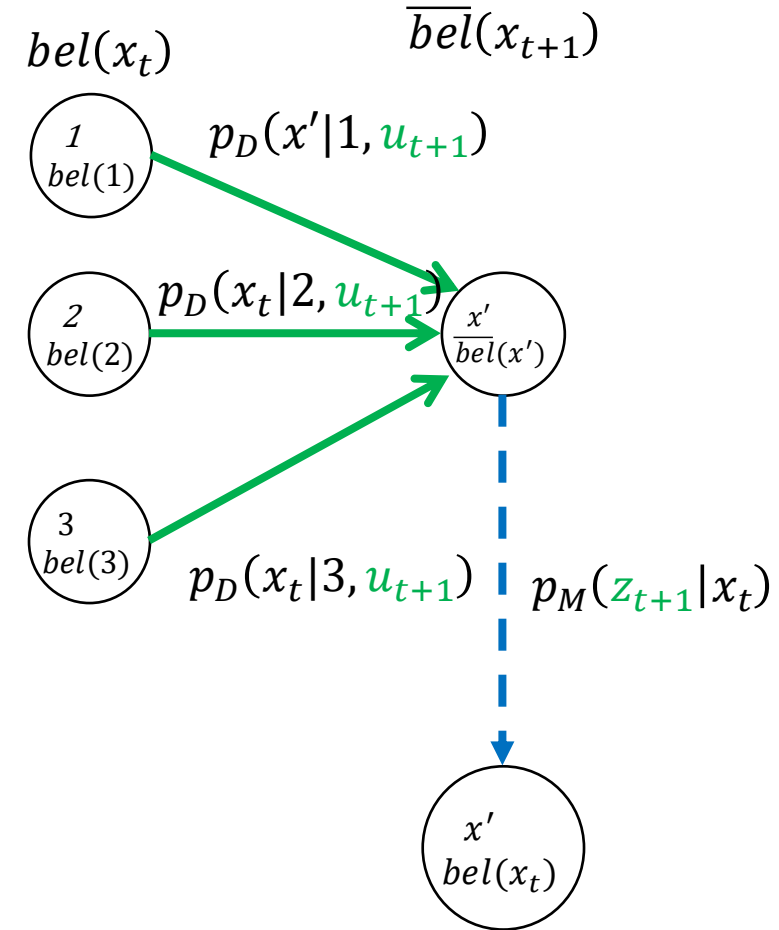
$$bel(x') = p(x' | z_{t+1}) = p(z_{t+1} | x') \cdot \frac{p(x')}{p(z_{t+1})} \quad \text{Bayes rule}$$

$$bel(x') = \eta p_M(Z_t = z_{t+1} | X_t = x) \overline{bel}(X_t = x)$$

where η is a normalizing constant to make $bel(x_{t+1}) \in \mathbf{P}(Q)$

$\overline{bel}(X_t = x)$ predicted prior; believed state **before** measurement

$p_M(z_{t+1} | x')$ is the likelihood of this sensor reading given the state is x'



Histogram Filter or Discrete Bayes Filter

Finitely many states $x_i, x_k, \text{etc.}$ Random state vector X_t

$p_{k,t}$: belief at time t for state x_k ; discrete probability distribution

Algorithm Discrete_Bayes_filter($\{p_{k,t-1}\}, u_t, z_t$):

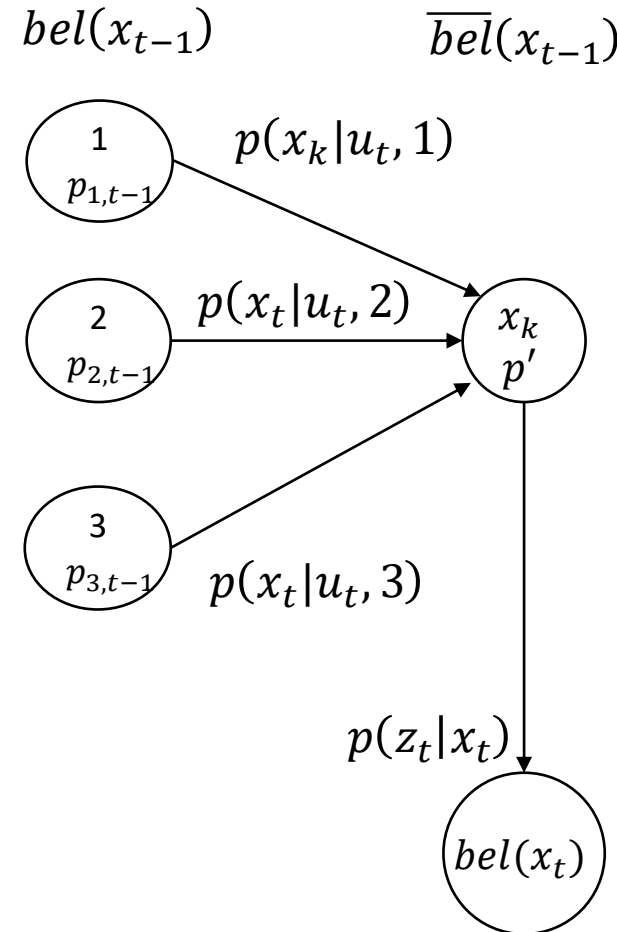
for all k do:

$$\bar{p}_{k,t} = \sum_i p(X_t = x_k | u_t, X_{t-1} = x_i) p_{i,t-1}$$

$$p_{k,t} = \eta p(z_t | X_t = x_k) \bar{p}_{k,t}$$

end for

return $\{p_{k,t}\}$



Grid localization

Algorithm Grid_localization ($\{p_{k,t-1}\}, u_t, z_t, m$)

for all k do:

$$\bar{p}_{k,t} = \sum_i p_{i,t-1} \mathbf{motion_model}(\text{mean}(x_k), u_t, \text{mean}(x_i))$$

$$p_{k,t} = \eta \bar{p}_{k,t} \mathbf{measurement_model}(z_t, \text{mean}(x_k), m)$$

end for

return $bel(x_t)$